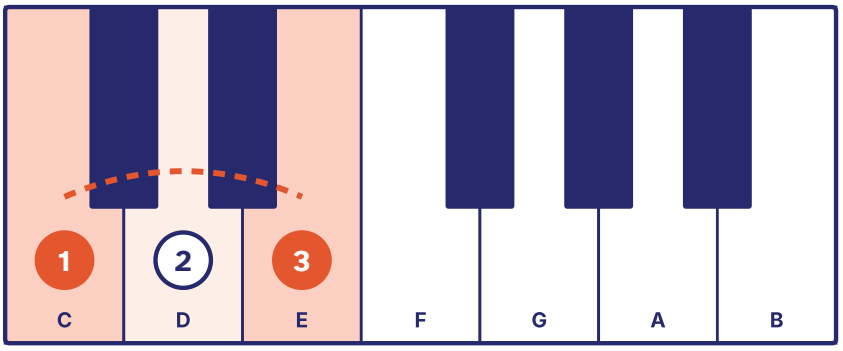
PART I · PITCH · 6

Counting intervals

How far apart are two notes? That distance is an **interval**. To give it a number, **count the notes**: the first note is 1, and you count every letter up to the second, both ends included.

THE SECRET

Count the notes, both ends included: C D E is 3 notes, a 3rd.

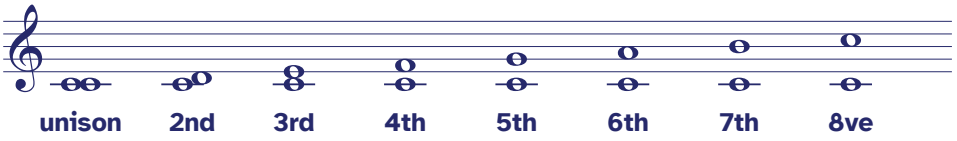


C to E: count the notes, 1 2 3: a 3rd

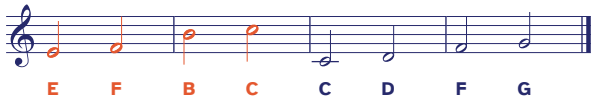
THE RULE

Count the notes for the number: start at 1 on the first note. C to E is C D E: 3 notes, a 3rd. Sharps and flats never change the number.

Interval numbers from C



The smallest steps: semitones and tones



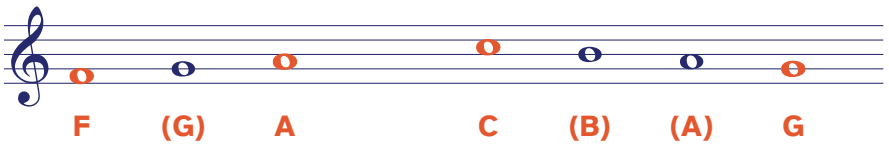
E–F and B–C: no black key between, a semitone. C–D and F–G: a black key between, a tone.

From any key to the **very next key**, black or white, is a **semitone**: the smallest step on the piano. Two semitones make a **tone**. On the white keys most steps are tones, but **E–F** and **B–C** have no black key between them: they are semitones.

PART I · PITCH · 6 · CONTINUED

EXAMPLE 1

How far apart are these notes?

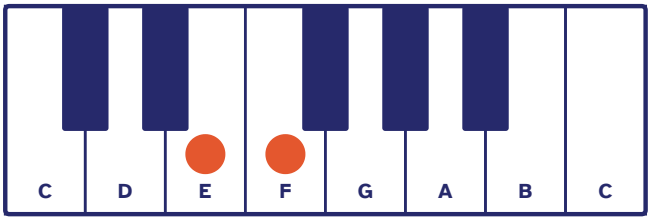


1. **Call the first note 1:** here F is 1.
2. **Count every letter** up to the second note.
3. **Going down works the same:** start at 1 on the top note and count down.

F G A is 3 notes: a **3rd**. C B A G is 4 notes: a **4th**.

EXAMPLE 2

Tone or semitone?

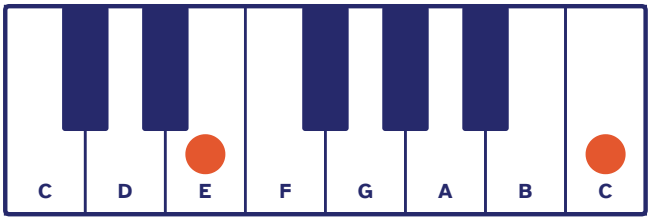


1. **Look between the red keys:** is there a black key?
2. **No black key** means they are next-door keys.
3. **Now try C and D**: is there a black key between?

E–F is a **semitone** (next door). C–D has C♯ between: a **tone**.

EXAMPLE 3

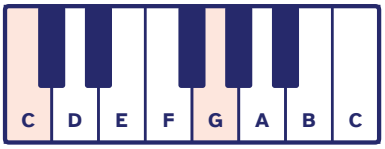
Name an interval on the keyboard



1. **Find both red keys:** E, and the C above it.
2. **Count the notes** from E: E is 1 …
3. **Stop on the C.** What number is it?

E F G A B C is 6 notes: a **6th**.

TRY IT · 10 SECONDS



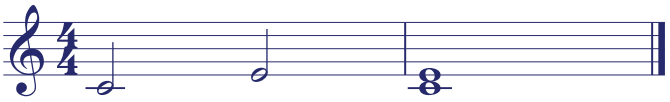
Thumb on C, finger 5 on G.

Your hand is a ruler: thumb to little finger on white keys is a 5th. Play it, then thumb to finger 3: a 3rd.

PART I · PITCH · 6 · CONTINUED

EXAMPLE 4

Melodic or harmonic?



1. **The first bar:** C, then E, one after the other.
2. **The second bar:** C and E at the same time.
3. **Count the notes** in each.

Both are a **3rd**: **melodic** when played one after the other, **harmonic** when played together.

GAME

Count the notes

1	2	3	4
C up to G	D up to F	E up to the next E	G up to A

Count every note, both ends included. What is each interval called?

Answer: 1 a 5th (C D E F G) · 2 a 3rd · 3 an octave (8) · 4 a 2nd.

YOU KNOW THIS

Twinkle, Twinkle, Little Star



The leap from the first “twinkle” to the second is C up to G: C D E F G, **five notes, a 5th**.

Try it. Sing the first two notes of any song, then count the notes between them.

THINK OF IT Monday to Wednesday is three days, not two: you count the day you start on. Intervals count the same way.

TRAP

- × Counting only the gaps, not the notes: C to E is 3 notes, not 2 gaps.
- × Letting a sharp or flat change the number: C to G♯ is still a 5th.

SEE ALSO

Major, minor and perfect p. 41 · Major scales p. 70

PRACTISE IT

Workbook Pitch 4 · Online: Library › Intervals

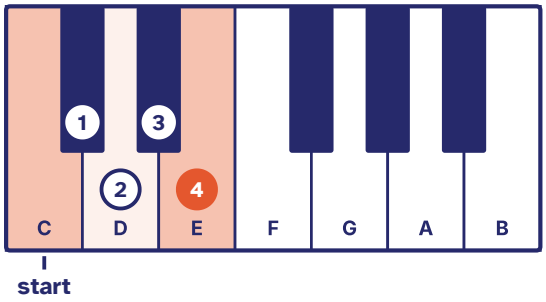
PART I · PITCH · 7

Major, minor and perfect

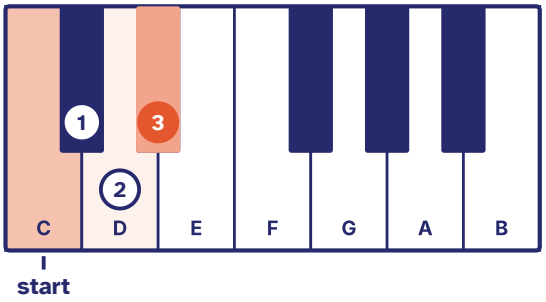
C to E and C to E♭ are both 3rds: three notes each. But they sound different, bright and dark. To tell them apart, **count the hops**: every move to the very next key, black or white, is one hop.

THE SECRET

Count the hops: 4 make a major 3rd, 3 a minor 3rd.



C to **E**: 4 hops. A **major** 3rd: bright.



C to **E♭**: 3 hops, one fewer. A **minor** 3rd: dark.

One hop less: a major 3rd becomes minor

THE RULE

Count the notes for the number (a 3rd). Count the hops for the size (major or minor). Do not count the key you start on: the first move is hop 1.

NUMBER	SMALLER	MAIN SIZE
3rd	3 (minor)	4 (major)
4th		5 (perfect)
5th		7 (perfect)
6th	8 (minor)	9 (major)
octave		12 (perfect)

3rds and 6ths come in two sizes: **major**, and **minor**, one hop smaller.

4ths, 5ths and octaves have one main size, called **perfect**. They are never major or minor.

EXAMPLE 1

Major or minor 3rd?



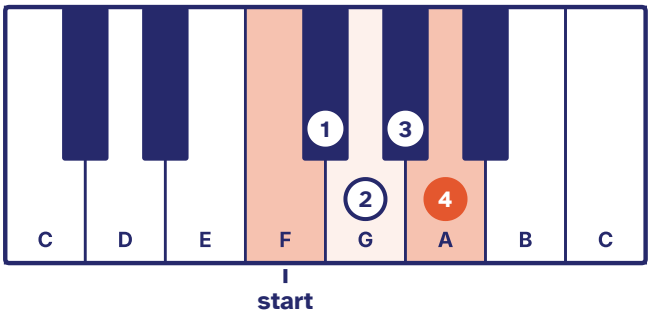
1. **Count the notes:** C D E, three letters, so it is a 3rd.
2. **Count the hops** from C to the red black key, E♭: C♯ 1, D 2 …
3. **Major has 4 hops.** How many here?

C to E♭ is **3 hops**: a **minor 3rd**.

PART I · PITCH · 7 · CONTINUED

EXAMPLE 2

Name F up to A

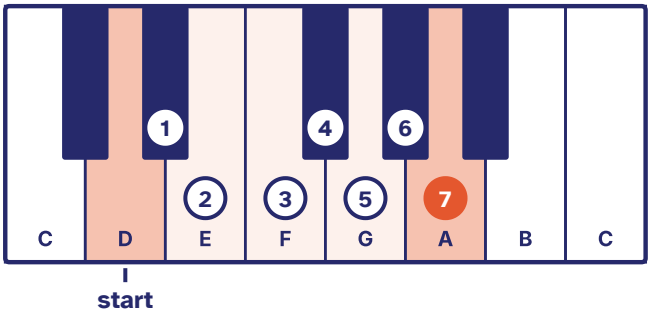


1. **Count the notes:** F G A is 3, so it is a 3rd.
2. **Count the hops:** every key you land on, black and white, is numbered.
3. **4 hops or 3?**

F to A is 4 hops: a **major 3rd**, the same size as C to E.

EXAMPLE 3

Name D up to A

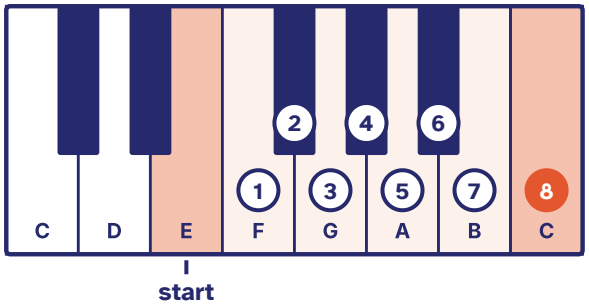


1. **The number:** D E F G A, five notes, a 5th.
2. **The size:** count the hops: the numbered keys.
3. **A 5th of 7 hops** is its main size. What is that called?

D up to A is a **perfect 5th**: 7 hops.

EXAMPLE 4

Name E up to C



1. **The number:** E F G A B C, six notes, a 6th.
2. **The size:** count the hops.
3. **A major 6th** has 9 hops. Is this one more or one fewer?

E up to C is 8 hops, one fewer than major: a **minor 6th**.

PART I · PITCH · 7 · CONTINUED

GAME

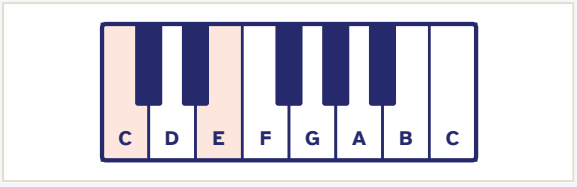
Perfect, major or minor?

1	2	3	4
C–G	D–F	C–E	E–A

Count the notes for the number, then the hops for the size.

Answer: 1 perfect 5th (7 hops) · 2 minor 3rd (3) · 3 major 3rd (4) · 4 perfect 4th (5).

TRY IT · 10 SECONDS

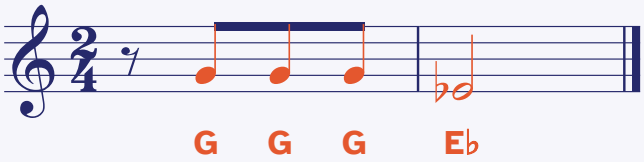


Play C–E, then C–E♭.

Bright, then dark. One hop makes the difference: the major 3rd has four, the minor 3rd three.

YOU KNOW THIS

Beethoven, Symphony No. 5



Da-da-da-DUM: G down to **E♭** is three notes, a 3rd, and 4 hops (F♯, F, E, E♭): a **major 3rd**.

Try it. Count the hops from G down to E♭ on the keyboard.

THINK OF IT A major and a minor 3rd are a smile and a frown: the same face, one small change at the corners.

TRAP

- × Counting white keys only: count every hop, black and white.
- × Counting the start key as hop 1: the first move is hop 1.
- × Calling a 4th or 5th “major”: they are perfect.

SEE ALSO

Counting intervals p. 38 · Chords and triads p. 87

PRACTISE IT

Workbook Pitch 4 · Online: Library › Intervals